

<Environmental Microorganism Image Analysis Using Deep Learning>

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Abstract—This study proposes an automated deep learning pipeline for environmental microorganism image analysis using EMDS-6. Images are denoised by a Two-Dimensional Rank Order Filter and DnCNN, then segmented by U-Net variants and DeepLabV3+ (U - Net++ achieving IoU 0.9365 and Dice 0.9843). Extracted RGB/HSV histograms and LBP textures enable accurate classification. The modular framework removes manual preprocessing, delivering robust, scalable microbial identification. Additionally, classification and object extraction modules are integrated, with ResNet18 achieving an F1-score of 0.9464 for microorganism classification.

Keywords—Environmental Microorganisms, Deep Learning, Image Denoising, Image Segmentation, Feature Extraction, Image Classification, Computer Vision

I. INTRODUCTION

Environmental microorganisms (EM) are one of the components of the ecological balance. Applications in public health, environmental monitoring, and industrial production are just a few of the affected areas. But traditional analysis methods based on manual microscopy are labor-intensive, time-consuming and subject to errors of human subjectivity. This project focuses on using deep learning to automate the analysis of EM images.

This project aims to design a system that streamlines and enhance important image analysis functions. Some of those tasks can be denoising, segmentation, feature extraction, and classification. The EMDS-6 dataset is used in this project. It focuses on offering a great framework. It aims to provide an effective framework that enhances analytical performance and reduces overall processing time for the detection of microbes. This innovative approach significantly improves efficiency compared to traditional manual analysis methods. We expect this innovation will revolutionize the way it is done now. Monitoring environmental quality, preventing outbreaks of disease, and enhancing industrial control are made easy.

II. LITERATURE REVIEW

Deep learning has seen a rapid growth of applications in the image processing area in the last years. This is especially imperative in areas like biomedical imaging and environmental monitoring. In contrast with the previous studies, which are primarily related to the process of denoising or segmentation, and its methods are predominantly based on classical algorithms or hand-crafted features. to handle the complexity of image data in the real world.

In recent years, the use of deep learning to detect microorganisms has become an important topic. In most studies, a combination of machine learning and deep learning methods is required. Zou et al. [1] proposed an environmental microbial image retrieval method based on multi-color channel fusion and particle swarm optimization. Saeid et al.

[2] proposed an algorithm that combines fuzzy C-means, SOM and LVQ neural network to segment and identify clusters. Osman et al. [3] used Zernike moments and a hybrid multilayer perceptron network to detect Mycobacterium tuberculosis in images. Massana et al. [4] discussed the preprocessing edge detection method of bacterial images under epifluorescence conditions. Computer software machines such as MB-Bact and ELISA [5] are used to detect specific types of bacteria, but they are expensive for those with weaker detection capabilities. Frank and his team [6] proposed a novel pairwise deep learning feature (PDLF) to analyze microorganisms and used support vector machine (SVM), linear discriminant analysis, logistic regression, XGBoost and random forest classifiers to classify EM. Based on the previous work, this project strives to design a deep learning framework that can show stronger adaptability and stability in practical applications.

III. METHODS

3.1.1.SPN+TROF

In our project, we use the 'TROF' (Two-Dimensional Rank Order Filter), which uses nonlinear local order statistical methods to eliminate salt-and-pepper noise from images and maintain structural characteristics. The performance evaluation of TROF shows superior results when compared to GMF and MinF for reducing image noise while maintaining details on EMDS-6.[7]

3.1.2. DnCNN

The image denoising task uses DnCNN as its basic residual convolutional neural network structure. This method predicts image noise and then uses this prediction to remove it from the input to restore a clear image. Essentially, by combining residual learning with batch normalization, fast convergence and robust denoising performance are achieved. [8] The experimental application of DnCNN has achieved success in grayscale microscopy images and produced competitive results.

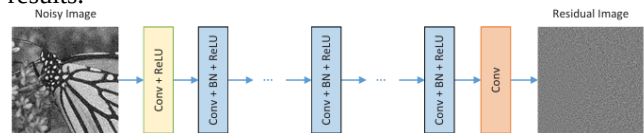


Figure1. The architecture of the proposed DnCNN network.[8]

3.2.1.U-Net

The U-Net architecture serves as a popular encoder-decoder method particularly optimized for image segmentation tasks through its implemented skip connections which maintain spatial information. The network is widely used in biological tasks due to its simplicity and strong performance [9].

3.2.2. DeepLabV3+

A semantic segmentation framework was implemented with DeepLabV3+ architecture. The research uses ResNet50 and ResNet101 backbone network models with pretrained weights downloaded from the ImageNet database.[10]

3.2.3. U-Net++

The enhanced version of U-Net called U-Net++ uses dense skip connections with nested decoders that lead to improved boundary detection alongside multi-scale representation. [11]. Early stopping in the training process allowed for preventing overfitting by implementing a strategy that monitored validation loss until no additional improvements appeared [12].

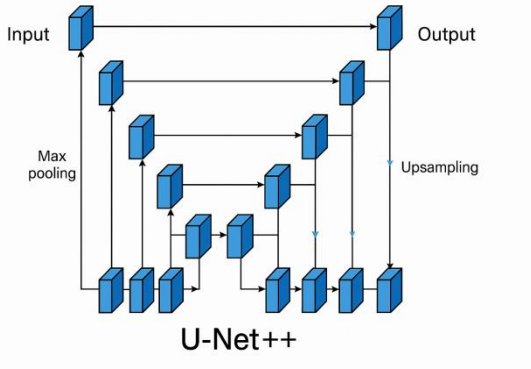


Figure2. Process of U-Net++

3.3.1. Feature extraction of RGB

We obtained RGB histogram data points from predicted foreground area boundaries of images through a binary mask application. We analyzed color distributions from white mask areas because these regions contained object information apart from background noise.[13] The normalization process creates vectors which depict shaped color intensity sections over R, G and B channels and function as classification inputs as proposed in.

3.3.2 Feature extraction of HSV

Images in the HSV color model display color attributes through hue, saturation and value channels. We obtain HSV histogram data exclusively from the target areas that are predicted to be colored objects (white mask areas) in order to minimize background distortions. This model demonstrates strong resistance to changes in illumination together with efficient color categorization which suits intricate observation fields including microscopy [14].

3.3.3 Feature extraction of LBP

The texture descriptor named Local Binary Pattern computes local structural information through pixel neighborhood threshold operations[15]. In our method, we first divide each 256×256 grayscale image into 8×8 non-overlapping blocks. Each block contains LBP calculation through 8 sampling points with a radius of 1 which generates its pattern histogram. The final outcome comprises normalized histograms obtained by concatenating both local pattern information with spatial distribution data.

3.4.1 Classification

In classification, this project uses Support Vector Machine (SVM) with RBF kernels and a various of deep learning

models, including convolutional neural networks (CNNs) such as ResNet18, ResNet34, ResNet50, EfficientNetB0, DenseNet169, Xception, DeiT and InceptionResNetV1, as well as transformer-based models including Vision Transformer (ViT) and a simplified version named SampleViT.

3.4.2 Cross Entropy

During training, we applied the cross-entropy loss function (1)

$$L_{CE} = - \sum_{i=1}^c y_i \cdot \log(\hat{y}_i) \quad (1)$$

where y_i is the ground truth label and $\hat{y}_i$ is the predicted probability for class i .

3.4.3 Adam Optimizer

We used the Adam optimizer to update model parameters as (2)

$$\theta \leftarrow \theta - \eta \cdot \frac{\hat{m}_t}{(\sqrt{\hat{v}_t + \epsilon})} \quad (2)$$

where $\hat{m}_t$ and $\hat{v}_t$ are the bias-corrected estimates of the first and second moments of the gradients.

3.5.1 R-CNN with ResNet-50

For object detection, there is a model based on Faster R-CNN with a ResNet-50 backbone, which pretrained on ImageNet.

3.5.2 SGD optimizer

The model was trained using the Stochastic Gradient Descent (SGD) optimizer with momentum as (3)

$$\theta \leftarrow \theta - \eta \cdot \nabla_{\theta} L + \mu \cdot v \quad (3)$$

where θ represents model parameters, η is the learning rate, μ is the momentum factor, and v is the velocity term from the previous update.

3.5.3 Loss in R-CNN with ResNet-50

The loss function in (4)

$$L = L_{CE} + \lambda \cdot L_{box} \quad (4)$$

Where λ is a coefficient to balance weight (usually 1). This loss function combines cross entropy and bounding box regression.

3.5.4 Box Regression Loss

As (5)

$$L_{box} = \begin{cases} 0.5 * (\hat{y}_i - y)^2, & \text{if } |\hat{y}_i - y| < 1 \\ |\hat{y}_i - y| - 0.5, & \text{if otherwise} \end{cases} \quad (5)$$

where $\hat{y}_i$ is predicted value and y is truth value. When $|\hat{y}_i - y| < 1$, the squared term $0.5 * (\hat{y}_i - y)^2$ is used — smoothly penalizing small deviations. In contrast, When $|\hat{y}_i - y| \geq 1$, the linear term $|\hat{y}_i - y| - 0.5$, is used — avoiding excessive penalty for large errors.

IV. EXPERIMENTS

The analysis utilized EMDS-6 dataset that consists of 840 grayscale electron microscope images divided into 21 categories which include pixel-level ground truth information([URL:https://figshare.com/articles/dataset/EMD_S6/17125025/1](https://figshare.com/articles/dataset/EMD_S6/17125025/1)). All images received two transformations: first a resize operation to 256×256 pixels followed by

normalization. The research evaluated three segmentation models (U-Net, U-Net++ DeepLabV3+) by training them from head using 80/20 data splits with Adam optimizer and early stopping and learning rate of 1e-4 and batch size of 4. The evaluation of segmentation quality used Dice coefficient alongside IoU and confusion matrix. The U-Net++ model achieved the greatest performance and was used for feature extraction from denoised DnCNN data.

For classification, this project firstly tested traditional handcrafted features with SVM classifiers. First, using denoised grayscale images extracted Local Binary Pattern (LBP) histograms, which helped capture texture information. Then, the project also extracted color histograms by inputting RGB and HSV images. To focus on significant regions, we used predicted segmentation results and computed color histograms. All feature vectors were normalized and used to train SVM models with RBF kernel (C=1 for LBP, C=20 for color features). In addition, the project also tested ResNet(18, 34, 50), Xception, DenseNet-169, InceptionResNetV1, DeiT, ViT, and SimpleViT. All models were fine-tuned from ImageNet pre-trained weights, trained with an 80/20 split (train/test), batch size of 32, and learning rate (0.0004) using Adam optimizer. The cross-entropy loss was also adopted. Models were trained on dataset, which consists of segmented regions overlaid on the original RGB images after being processed by U-Net++. Finally, the project estimated performance evaluation across preprocessing strategies, Metrics include accuracy, F1-score, and confusion matrix.

To perform object detection, this project implemented Faster R-CNN with a ResNet-50 backbone, pretrained on ImageNet. The model was fine-tuned on bounding boxes generated from segmentation images, searching detection targets. The optimizer was SGD with momentum of 0.9 and initial learning rate of 0.001. The loss function combined cross entropy (1) and Smooth L1 bounding box regression loss(5). As a result, Faster R-CNN is able to localize microbes with acceptable precision under microscope image conditions.

V. RESULTS

5.1

Our project used three different methods to split two datasets, first generating a confusion matrix, and then using Dice and IOU to determine which one has the best performance.

The formula for Dice is:

$$\frac{2TP}{2TP + FP + FN} \quad (6)$$

The formula for IOU is:

$$\frac{TP}{TP + FP + FN} \quad (7)$$

Table 1. Comparison of Segmentation Performance Across Models and Datasets

Model	IOU (SPN_0.03)	IOU (DnCNN)	Dice (SPN_0.03)	Dice (DnCNN)
U-Net	0.8527	0.8645	0.9133	0.9224
U-Net++	0.9160	0.9365	0.9642	0.9843

Model	IOU (SPN_0.03)	IOU (DnCNN)	Dice (SPN_0.03)	Dice (DnCNN)
DeepLabV3+	0.8559	0.8750	0.9587	0.8685

Based on the comparison of the results, we have decided to continue the project using U-Net++ to process the denoised images in the following steps.

5.2

This project compared some performance of different traditional features using SVM in classification. The LBP-based SVM got 26.2% accuracy and a macro F1-score of 0.21, the RGB-based SVM performed better with 38% accuracy and a macro F1-score of 0.37 and the HSV-based SVM got 32.0% accuracy and a macro F1-score of 0.32. These findings suggest that color-based features capture more critical information than texture-based features like LBP in microorganism classification. However, it was much lower than CNN-based models, especially the deep features for complex image recognition tasks.

In deep learning models, ResNet18 is the highest classification performance with an F1-score of 0.9464, meanwhile followed by Xception and DenseNet-169. Transformer-based models like ViT and DeiT underperformed due to limited training data. The results are shown in Table 2. The identical values of accuracy, F1-score result from the use of micro-averaging in torchmetrics, which makes same results when all classes are trained well in globally. The reason why the model CNN-based is better than the Transformer-based is likely that, the EMDS6 dataset is limited, and the local feature is an advantages of CNNs in small-datasets.

Table 2. Comparison of Classification Performance

Model	Best Val Accuracy (%)	F1-score
ResNet18	94.64	0.9464
Legacy Xception	88.10	0.8622
DenseNet-169	85.71	0.8502
ResNet50	83.33	0.8333
EfficientNetB0	80.36	0.8036
ResNet-34	75.60	0.7494
InceptionResNetV1	72.62	0.5737
SimpleViT	47.02	0.4702
ViT	21.43	0.2143
DeiT	28.57	0.2048

Here the formula to calculate Accuracy is (8) as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

The F1-score is (9) as:

$$F1 - score = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \quad (9)$$

In particularly, the precision in (8) is (10) as:

$$Precision = \frac{TP}{TP + FP} \quad (10)$$

And also the recall in (8) is (11) as:

$$Recall = \frac{TP}{TP + FN} \quad (11)$$

5.3

The loss showed a declining trend, indicating stable and effective training of Faster R-CNN. The final loss converged close 0.03, showing reliable capabilities of object detection. The loss in each epoch as table 3.

Table 3. Loss range of Object detection

Epoch	Loss Range
1	3.51 → 0.20
2	0.19 → 0.09
3	0.18 → 0.05
4-10	0.15 → 0.03

VI. CONCLUSION

In this study, a sequential method for completion of denoising, segmentation, feature extraction and classification tasks of environmental microbial images was established. The experimental verification results based on the EMDS-6 dataset were good. This eliminates the need for manual preprocessing and manual feature engineering and replaces it with deep neural networks. It is still able to produce stable output direction over varying sample types and large amounts of varying noise conditions. Also, all submodules are interchangeable and upgradable independently. It is convenient to replace new technologies in the future. It improves the flexibility and maintainability of the system.

However, this study still has the following limitations despite the above succeeds. One thing is that the data scale is small. In EMDS-6, this dataset contains only 40 images for each category. We need to improve the generalization ability of few-sample categories. The optimized for performance Single image takes are much longer for inference. As a result, it is hard to satisfy online monitoring needs in large-scale with low latency. Given the limitations, future directions for improvement may be as follows. As an example, the few-sample genres can be adjusted using generative adversarial networks (GAN).

Overall, this study verified the effectiveness of deep learning in environmental microbial image analysis. At the same time, it provides a clear direction for subsequent optimization and practical application.

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